



Amka Lighting's mid bay/high bay LED light

and therefore the fixing of lenses in the lights plays an important role. By using perfect lenses, the perfect spread of the light can be achieved."

In commercial lighting, design of the lighting fixture should be according to IP standards. If the life of the LED is more than 50,000 hours or 100,000 hours, the standard of the fixture should also match that.

"Many inferior designs are available in the market where the life of the fixture cannot be expected to extend beyond three to four years. How can we then expect to protect the LEDs for about 15 years with these fixtures? At Olive, we take special care of this aspect and use a good-quality metal to match the heat-sink quality," shares Jindal.

With drastic improvements in diffusers, secondary lens and reflector technology, it is now possible to achieve better-quality lighting with a good spread of light, without much wastage of energy or multiple shadows.



Binay's COB LED footlight

"Earlier, even though we had LEDs which could have been driven at up to 1 amp (3 watts), the lumen output was not efficient. But now with new-generation LEDs, the lumen output is much more. For example, even at 1 watt (350 mA) we get 130 lumens, or at 3 watts (1 amp) we get 300 lumens. This will help us to use fewer LEDs," says Gururaja.

Sujana Energy uses an intelligent proximity module such as a motion sensor in its products. The sensor is sensitive to movement and can offer functionalities like auto-dimming and auto switch-on/off. The company also offers photosensitive LED billboard lighting, which is solar-based and has an inbuilt dimming and on/off mechanism.

Nichia recently introduced low-power LEDs in tiny packages. The smallest LEDs ($3 \times 1.4 \text{ mm}^2$) now have inbuilt zener diodes, with the same footprint for 0.06-watt, 0.12-watt, 0.24-watt and 0.5-watt bulbs.

"Chip-on-board (COB) is an emerging technology which allows for virtually limitless freedom in the use of PCBs and thus serves as the basis for totally unique LED solutions," informs Ravi A.V.

COB technology is used for both small and high-power chips to give long life and high reliability to the product. In COB production process, LED chips are directly attached to the metal PCB substrate so that the heat generated at the p-n junction has a direct path for maximum heat dissipation.

"COB technology, though slightly expensive, is the best solution for temperature-sensitive devices such as LEDs, primarily due to its superior heat dissipation characteristics which result in cooler LED chips and a long life of up to 100,000 hours," explains Rohtagi.

"Our COB LED strips are standard products and we are also using COB technology to manufacture COB LED downlights, COB LED footlights, COB retrofit tubelights and COB retrofit GLS bulbs," says Rohtagi.

"Samsung's 11W COB solution provides up to 118 lumens/watt at

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5000K and 88-90 lumens/watt at 2700K. Thermals are adequately taken care of since the chip itself has a large